Arithmetic Chaos with Rare Islands of Order: A Computational Map of the Collatz Space

Part I: The Computational Map

Collatz Crystal Hunter Project

August 2026

Abstract

We present a large-scale computational study of Collatz trajectories in terms of accelerated odd dynamics, shift-vectors, and confluence structures, up to ~ 310 bits and trajectory lengths $d \approx 500$. The central discovery is **Zone 2** — a family of 913 numbers (bit lengths 71–87) whose trajectories merge into a single 75-bit node $x^* = 20152090995747160937051$ within ≤ 7 odd steps and then follow an identical path to a peak of 140 bits. In parallel, Barina's number is an isolated second path to the same peak. A systematic census reveals confluence centers for all integer peaks 14–51 plus peak 140 (39 centers), forming a continuous archipelago. We find the scaling law $\text{bits}(c) = 0.498258 \cdot \text{peak} + 6.2928$ ($R^2 = 0.965$, primary centers) and the modular filter $c \equiv 2 \pmod 3$. Cluster analysis separates **Class A** (deep funnels: 121, x^*; 100% hit rate, $S/d \approx 1.33$) from **Class B** (shallow confluences, 72–93%). A dead zone (88–170 bits) free of ratio anomalies is confirmed by four independent methods. The body of results indicates that the Collatz space is a continuous archipelago of confluence structures (**Class B**) above which rise rare deep funnels (**Class A**) organized around $\log_2 3$. This is Part I of a two-part programme; the companion Part II supplies the large-deviation theory explaining these structures. *Companion paper: Crystals Cannot Be Grown: A Large-Deviation Theory of Typical and Rigid Collatz Structures* (Part II).

Glossary and Constants

- **Family A ($2^b - 1$)**: base landscape; ratios $\to \log_2 3 \approx 1.585$.
- **Confluence**: capture of trajectories into nodes (centers).
- **Class A (deep funnels)**: 100% hit rate; centers 121 (peak 14), x^* (peak 140).
- **Class B (shallow funnels)**: 72–93% hit rate; dense archipelago (peaks 14–51).
- **Center-bits formula**: $\text{bits}(c) = 0.498258\,\text{peak} + 6.2928$ ($R^2 = 0.965$); refit over all 39 centers $0.620145\,\text{peak} + 2.2850$ ($R^2 = 0.916$).
- **Scaling Hypothesis** $\times 10$: Class A centers follow a logarithmic scale ($14 \to 140 \to 1400$); constructive routes closed (Part I §7); existence open.
- **Septembrino's Law**: divisor distribution $P(\text{div} = 2^a) = 2^{-a}$.

1 Introduction and Chronology

1.1 The Collatz problem

For $n \in \mathbb{N}$ define $T(n) = n/2$ (even) and $T(n) = (3n+1)/2$ (odd). The Collatz conjecture states that iterating T from any n reaches 1. The anomaly of a number is its *ratio* (peak bits / input bits); typical numbers give ratio ≈ 1.0–1.2; Family A gives ratio $\to \log_2 3$.

1.2 Starting point: Barina's records

Barina's exhaustive search to 2^{71} lists two 71-bit record holders with peak 140; these seeded the study.

1.3 Phases

Phase 1 (Markov search, 35 cores): best ratio 1.613 for 73–80 bits. **Phase 2** (30M records): ratio peak ≈ 1.04; max-ratio$\times$bits recomputed per bit length always equals 140 (the "magnetic peak"). **Phase 3** (structural): CRT constructor, shift-vector analyzer, reverse trees, beam search; all results below obtained.

2 Notation and Accelerated Dynamics

We use accelerated odd dynamics $x_{k+1} = (3x_k + 1)/2^{a_k}$ with $a_k \geq 1$; the shift-vector is $(a_0, \ldots, a_{d-1})$; $S_k = \sum_{i<k} a_i$; after k steps $x_k = (3^k x_0 + c_k)/2^{S_k}$, $c_0 = 0$, $c_{k+1} = 3c_k + 2^{S_k}$. The cumulative gain is $G(k) = k \log_2 3 - S_k$; the peak is near $\max G$. Every prefix defines a residue class $x_0 \equiv r_k \pmod{2^{S_k}}$.

2.1 2-adic interpretation

An admissible word w defines a cylinder $C(w) = \rho(w) + 2^{S_d}\mathbb{Z}_2$ of Haar measure 2^{-S_d}. **Zone 2** is a cylinder of measure 2^{-342}; 27 gives 2^{-70}; Barina gives 2^{-271}. The archipelago is a finite set of exceptional cylinders separated by vast regions of exponentially small growing-path measure, consistent with Tao's almost-all decay.

Remark 1 (Septembrino's parameterization). $N = k \cdot 3^m - 1$ parameterizes the Haar measure in $\mathbb{Z}_2$; the observed periodicity of $v_2(N)$ follows from $3^n \bmod 2^p$, placing it in classical p-adic dynamics.

3 Base Layer: Family A

Numbers $2^b - 1$ have shift-vectors 94–97% ones, ratio $\to \log_2 3$, $S/d \approx 0.99$ — the landscape level. Full spectral analysis $b = 71$–310 found 10 values with $S/d > 1.05$ (micro-plateaus at peaks $183, 190, 280, 483, 494$), all resonances of $\log_2 3$ continued-fraction approximations; verification showed micro-plateaus exist only for $2^b - 1$ itself.

4 Zone 2: The Only Confirmed Major Confluence Class

4.1 Discovery and representatives

Barina record holders (71–82 bits) all peak at 140 with shift-vectors identical from position 8; **Zone 2** found for all bits 71–87.

Bits	**Number** n	**Peak**	**Ratio**	**Steps to** x^*
71	2358909599867980429759	140	1.9718	7
75	37742553597887686876149	140	1.8667	7
80	1207761715132405980037043	140	1.7500	7
87	154593499536947965444748439	140	1.6092	7

Representatives (full 17-row table in Supplementary). All have $d = 259$, $S = \text{bits} + 271$.

$2^{88} - 1$ peaks at 141 and does *not* pass through x^* — the sharp boundary.

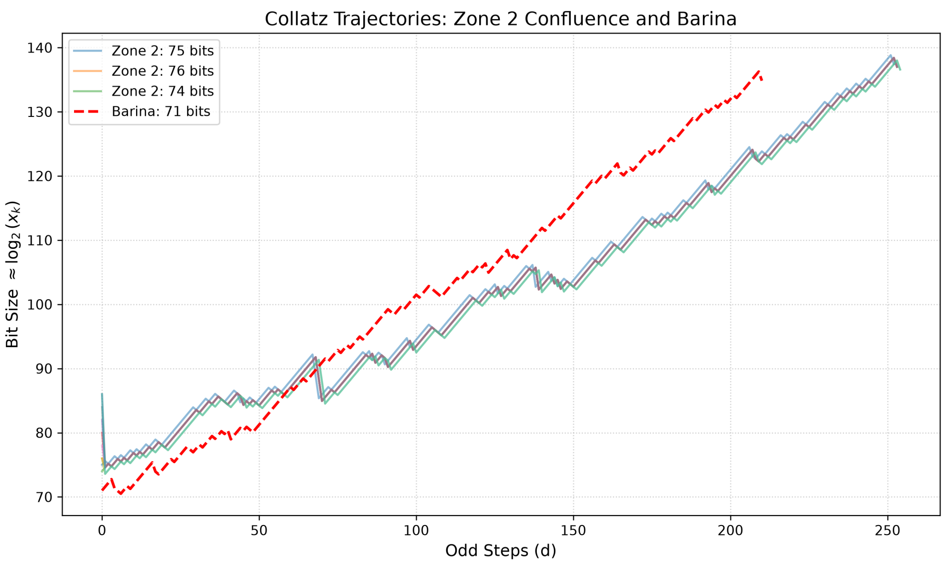

Figure 1: Trajectories of Zone 2 numbers (solid) perfectly merging into x^*, compared to Barina's number (dashed) reaching the same peak independently.

4.2 Two-layer invariants

Core ($x^* \to$ peak): identical for all 913; $d_{core} = 251$, $S_{core} = 334$, $S/d \approx 1.331$. **Adapter shell**: inputs reach x^* in $k \in [0, 7]$ steps; outer shell ($k = 7$) obeys $S_{adapter} = \text{bits} - 63$; adapter entropy ≈ 0.395 bits^{-1}; adapter determined by lowest 19 bits.

4.3 Confluence point x^*

Theorem 2 (Computational)**.** *All 17 representatives (bits 71–87, $d = 259$) arrive at $x^* = 2015209099574716093705$1 (75 bits) in ≤ 7 odd steps; trajectories then identical (exact equality) for 252 steps to peak 140. $2^{88} - 1$ does not pass x^*.*

4.4 Reverse tree and full size

Reverse tree of x^* to depth 7 (bits ≤ 90): 1859 nodes, 913 with bits 71–87; all 913 yield peak 140 (100%). Inputs double per bit.

4.5 Grammar and the $(1,1,2)$ equilibrium

FFT shows periods $T \approx 10, 6.3, 18$; critical tail shows staircase $[2,1,1]$, the shadow of the 3-adic point $-29/11$.

Lemma 3 ($(1,1,2)$ periodic equilibrium). *$R_a(x) = (x2^a - 1)/3$; $F = R_2 \circ R_1 \circ R_1 = (16x - 29)/27$, a contraction on $\mathbb{Z}_2$ ($|16/27|_2 = 2^{-4}$) with fixed point $\xi = -29/11$; mean shift $4/3$, gain $\log_2 3 - 4/3 \approx 0.2516$, matching body growth ~ 0.25.*

The core satisfies $S/d \approx 1.331$ (within 0.2% of 4/3): the core is the vacuum ξ dressed by a sparse defect gas. (Full instanton analysis: Part II.)

5 Barina's Number: An Isolated Second Path

$n = 1765856170146672440559$ (71 bits, $d = 213$, $S/d = 1.263$) reaches peak 140 via a path sharing no intermediate point with Zone 2 (even mod 2^{30}); common prefix $[1,1,1]$; reverse tree from Barina's peak empty (peak odd value $\equiv 0 \pmod 3$, no predecessors). Same peak, two independent mechanisms.

6 Number 27 as a Miniature Zone 2

27: $d_{peak} = 28$, $S/d \approx 1.36$, shift profile near-identical to Zone 2; $x_7 = 121$ is a confluence center (reverse tree depth 5: 27 inputs, 100% peak 14); similarity score $\{x^*, 27\}$ vs $\{\text{Barina}\} \approx 0.989$.

7 Dead Zone 88–170 Bits

No ratio > 1.585 found in 88–170 bits by four methods (peak hunter, parity scan $>$ 14M checks, beam search $\pm$ niching). **Corollary.** After Zone 2 (boundary 87 bits) no new confluence *ratio* anomaly to 170 bits.

7.1 Probabilistic anatomy (Sanov)

Under the i.i.d. shift heuristic, $P(S/d \leq 1.40) \sim 2^{-d D_{\mathrm{KL}}(q\|p)}$, $D_{\mathrm{KL}} \approx 0.28$; for $d > 300$, $< 10^{-25}$. (A heuristic, not a bound.)

7.2 Algebraic impossibility of shadow zones (CRT dimensionality)

Proposition 4 (CRT dimensionality trap). *An admissible block w of shift S repeated m-fold occupies one residue class mod 2^{mS}; for target bits $B < mS$ the expected count is $\approx 2^{B-mS}$.*

For Barina ($S \approx 269$) a 2-fold echo at $B = 142$ has expectation 2^{-396}; Zone 2 core ($S = 342$–358) gives $\leq 2^{-514}$; x^* gives $\leq 2^{-496}$. Constructive CRT lifts confirm echoes exist near $B \approx 2S$ (670/544 bits) but with ratios $1.19/1.25 < 1.585$: algebraically natural echoes fail to breach the large-deviation limit.

7.3 Control-budget obstruction

A defect $a \geq 3$ forces a clean run $L \geq (a - \bar{a}_t)/(\bar{a}_t - \bar{a}_p)$; with $\bar{a}_p = 4/3$, $L_{\min} \approx 82.5$–$99 \gg L_{\max} \approx 3.4$ for $B = 4$.

Corollary 5 (Closure of constructive Scaling $\times 10$)**.** *No bounded-branching constructive lifting assembles a Class A macro-soliton.*

Remark 6 (Existence vs. constructibility). The obstruction is algorithmic, not ontological: solitons can be found, not grown.

8 Refutation of Zone 3

Candidates (peaks 237–244, 145–150 bits) are Family A neighbourhood (density 1.000, bit-inversion fragility identical to Family A). Zone 3 does not exist.

9 Archipelago of Confluence Centers

9.1 Census (peaks 10–200)

Verification (reverse tree, depth 7) confirmed centers for all integer peaks 14–51 plus 140. Directed search (up to 5.7×10^9 candidates/peak) found centers for every peak 31–50. Recurring hubs (2051, 23) are *descent* transit hubs, not caustics (filtered out).

Peak	**Center**	**Bits**	**Inputs**	**HR**	d_{peak}	S/d
14	719	10	33	80.5%	3	1.00
14	121 (A)	7	27	100%	21	1.38
20	4763	13	—	100%	—	—
30	58595471	26	31	81.6%	3	1.00
40	35337455	26	343	91.7%	41	1.27
50	1396693151	31	716	88.5%	53	1.26
51	6572463707	33	≈ 400	88.0%	56	1.29
140	x^* (A)	75	913	100%	250	1.33

Selected rows; full 39-center table in Supplementary.

9.2 Two classes; continuum of caustics

Class A $\{121, x^*\}$: 100% HR, $S/d \approx 1.357$, compact (bits/peak ≈ 0.518), long path. Class B: 72–92%, $S/d \approx 1.252$. Depth-resolved HR shows Class B/transitional collapse by depth 4–5 while x^* holds $\geq 95.9\%$ to depth 9: Class A is singular; transitional centers are the strong end of Class B. Genuine ascending centers reach $HR = 1.0$ as the window deepens (local caustics); transit hubs dissolve ($HR \to 0$).

9.3 Centers do not cover the space

156/17000 (0.92%) random trajectories pass a center; 0/14 with ratio>1.5.

10 Spectrum of Chords

S/d is chord strength: Family A 0.99; weak 1.05; intermediate 1.19; Zone 2 (Barina) 1.26; Zone 2 core 1.32; 27 at 1.36 — a continuous spectrum, Zone 2 and 27 in the same range.

11 Summary Map

Family A base; continuous Class B archipelago (peaks 14–51); two Class A deep funnels; Zone 2 (913 inputs, x^*); Barina isolated; dead zone 88–170; plateaus 280/483 as Family A resonances; macro-centers non-constructible ($\Delta \geq 2P$).

12 The Big Picture

The space is a continuous field of confluence structures (Class B) above which rise rare deep funnels (Class A). Crystals can be found, not grown (triad of obstructions: Sanov, CRT dimensionality, control budget). Scaling $\times 10$ is a question of existence, not constructibility.

13 Computational Complexity

Exhaustive parity scan $\mathcal{O}(2^{bits})$ (infeasible > 90); reverse tree $\mathcal{O}(B^{depth})$ (Zone 2 in ms); residue beam search (to 170 bits); confluence census (22.9×10^9 candidates for peak 51).

14 Open Questions

Archipelago beyond peak 50; existence of x^{**} (peak 1400); nature of $\alpha = 1/2$; Barina's intermediate point; compatibility with Tao's theorem; $\mathbb{Z}_2$-corridor stability of Class A vs B. (Theoretical explanations of all the above are developed in Part II.)

Data Availability and Supplementary Material

Full Zone 2 catalog (913 entries), 39-center census, KL derivation, beam-search death logs, and algorithmic pseudocodes are in the Supplementary Material and the project GitHub repository (see README). All verification scripts (`zone_parity_search.py`, `reverse_tree_xstar.py`, `confluence_census.py`, `verify_census_centers.py`, `targeted_search_3` `two_class_analysis.py`, `basin_test.py`, `family_a_spectrum.py`, `automaton_invariants.py`, etc.) are released with the repository.

15 Conclusion: The Computational Foundation

This Part has constructed a complete computational map of the Collatz space to ~ 310 bits: Zone 2 (913 inputs, two-layer core+adapter), the continuous archipelago (39 centers, scaling law, two classes), the $(1, 1, 2)$ vacuum, the dead zone, and the triad of obstructions closing all constructive routes. **Philosophical conclusion.** The space is not "chaos with rare islands" but a continuous field of confluence structures above which rise rare deep funnels; these can be found but not grown. The theoretical explanation — the large-deviation duality, dressed-vacuum anatomy, and the two-heads dichotomy — is developed in the companion Part II.

References

[1] Lagarias, J.C. (1985). *The 3x+1 problem and its generalizations.* Amer. Math. Monthly 92(1), 3–23.

[2] Tao, T. (2022). *Almost all orbits of the Collatz map attain almost bounded values.* Forum Math. Sigma 10, e12.

[3] Barina, D. (2021). *Convergence verification of the Collatz problem.* J. Supercomputing 77, 2681–2688.

[4] Dembo, A., Zeitouni, O. (1998). *Large Deviations Techniques and Applications.* Springer.

[5] Collatz Crystal Hunter Project. *Crystals Cannot Be Grown: A Large-Deviation Theory of Typical and Rigid Collatz Structures* (Part II).